Binary is All You Need

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Abstract

The current development on radio technology shows an innovative approach in increasing the communication efficiency of several language internationally, while research in the related fields in terms of Frasia language was still blank. This article will propose a feasible protocol specially designed for Frasia language in digital binary format, with its corresponding mechanism, and to generate a Frasia system structure specially for the language transferring, which can replicate the 1544 case energy.

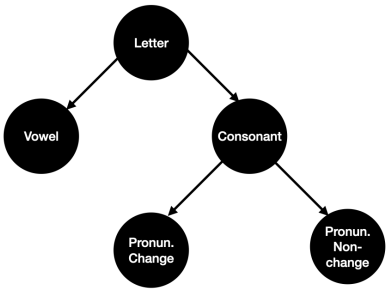

Figure 1: Binary Tree Decision

Notice

Please note that this paper is a secondary creation of Video Game *Knell of St. GodhrKar*, which is fully **not related** to any of the real society, organization, religion or country. The unknown languages and cases mentioned in this paper (e.g. Frasia) are all **self-created content** in the game, but perfectly conform to the binary form in current network, which inspired this paper's original idea.

1 Introduction

It is widely known that Frasia is a natural language found by musicians that can retrieve energy from *Andinomie* in the six-dimensional space [2], while the current interpretation is highly related to musical knowledge and perception. As a result, the language learning and development is always troublesome due to its specific pronunciation and deformation, especially for people who lack musical knowledge and are hard to tell the difference between pitch and notes. Meanwhile, musicians always do not have the appropriate knowledge of nowadays technologies, which makes the dissemination of Frasia lyrics still based on handwriting or even oral, where there is a high chance of human-made mistake occurring, and a simple mistake in lyrics or pronunciation will cause an uncontrollable disaster, which has been proved in history, bringing plenty of security risks to society [16].

We have noticed that scientists of the Verithesis Order have already done some research based on radio techniques to find an efficient way to broadcast [14] over long distances with little error. However, the system is not stable enough to fit our requirements when processing Frasia message as the broadcast is still audial. Therefore, we still need musicians to interpret the message after receiving the radio signal. However, the application of conversion from audio signal to electrical signal has activated us think about the digitization of Frasia language.

When we collect the electrical signal for radio generation, we always have a series of continuous signal that need to be shifted to modulation signal. Normally, the engineer will collect the sample periodically, and quantize each sample to a specific reading that has been pre-settled in a set that includes several electrical voltage. Intuitively, the set should be as big as enough to better simulate the original signal. However, consider an extreme case where we only have 2 pre-settled voltage inside the set, which is 0 and 1, which is perfectly fit our Frasia language characteristics because the reshaping of the Frasia letters can be considered as a binary tree 1. We have to decide the type of the letter, that whether it is vowel or consonant first, then decide the consonant pronunciation based on the tonality.

Following this idea, we can start to think about represent one single Frasia letter in several variables in binary, where each True or False represents one decision time (i.e. children branch). By considering the Frasia letter in binary form, the transportation of Frasia letters through electrical signal will not facing the noise issue that may affect accuracy. Further more, the protocol which is capable for Frasia letters transmitting process, and particular communication system that specially built for Frasia lan-

guage should be reviewed. Thus, this article is aim to deliver:

Digital Simulation: a base-2 mathematical interpretation of Frasia language in section 2;

Formatting: the proper form of Frasia message packets that include best size linear analysis, grammar, security design and communication mechanism in section 3;

Frasia System: a feasible and general network structure which allows the co-workings among tonalities and is able to produce the desired energy wave in section 4.

The reason that why we choose to deal with Frasia language instead of Andinomie language is that we have observed 3 times of significant revolution and 1 time evolution of Andinomie language after *The Explosion* [11], while the Frasia language still remains its original state same as the old era since it is highly related to natural phenomena. We have met the problem of losing language when centuries changed. Thus, we shall exert more effort on saving the most stable language, i.e. Frasia language, to support us on the communication and re-establishment. We expect this paper to provide a clear and easily-understand mathematical view of Frasia language to all the readers regardless the races they belongs to in a numerical way. We need to be aware that humanity should use this powerful language with sanity and wisdom, instead of devoting themselves to chaos and blindness. Although not every person is talented to use Frasia as what they expected, it may still be necessary for everyone to understand this language, not only for the harmony of society, but also for the acknowledgment of this interesting and paradoxical world.

2 Basic Approach

This section will propose a basic mathematical interpretation of the three common deformations of Frasia characters. Firstly, We need to summarize the common syntax rules used in Frasia language. Below are the definitions of the words that we are going to follow in this paper.

We give the definition of the three deformation types in Frasia as following[1]:

1. **Transposition:** This action initially refers to the pitch change of the whole scale when its tonality changes. In Frasia, this word refers to the scenario in which the pronunciation changes of all letters due to the tonality changing.
2. **Accidental:** This action initially refers to the pitch change of a single note when performing a music sheet. In Frasia, this word refers to the scenario in which the pronunciation changes of a single letter (not globally).
3. **Sustenance:** This action refers to keeping the Frasia letter type the same as the previous, either vowel or consonant.

We give the basic reading principles of Frasia as follows:

1. There are a total of 12 letters in Frasia, each representing 2 consonant and 1 vowel. Each letter at each time will have only one pronunciation. For example, Teyd cannot be read as T and D at the same time.
2. The pronunciation of consonant is determined by the existence of an accidental mark on current letter and the consonant pronunciation in the tonality itself. For example, if we have a letter Teyd in D major, it should be read as D without an accidental mark, and T with an accidental mark.
3. The reading sequence matters as well, basically, readers should spell Frasia in the sequence of consonant-vowel-consonant, or vise versa. If there is a sustenance remark on the current letter, its type should follow the previous letter's type. For example, when we spell the word *Frasia* in Frasia, obviously there are sustenance marks on letter r and letter a.

We will digitalize the three deformation types in section 2.1 and three reading principles in section 2.4.

2.1 Transposition of Consonants

In Frasia, we have two baseline tonalities defining the baseline consonant Frasia characters, which are C major (Tey) and A minor (Eyd). Take C major as an example, in this tonality, each consonant is appearing in their basic status, which are (sorted by scale):

$$Tey = [T, Th, H, Tr, M, F, S, P, C, L, Z, K] \tag{1}$$

In other word, this list can be viewed as a 1*12 matrix C,

$$C = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \tag{2}$$

where 1 represents the original non-changed status of each letter.

However, when it comes to D major, the matrix is changed because the pitches change. The D scale (Ha) consists of D, E, F-sharp, G, A, B, and C-sharp. We can easily see that notes F and C have been changed to fit in the major scale. Thus, in Frasia, the characters teyd (C), fyv (F), thh (C-sharp), and siu (F-sharp, also as G-flat) will be changed. From which we can have:

$$Ha = [D, Dh, H, Tr.M, V, Sh, P, C, L, Z, K] \tag{3}$$

which can be viewed as a 1*12 list D where 1 represents the non-change status and 0 represent the changing of each alphabet:

$$D = \begin{bmatrix} 0 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \tag{4}$$

We can easily see that the logic used to calculate these values in D scale is as follows.

$$T[i] = \begin{cases} 1, & \text{if X[i] = Tey[i]} \\ 0, & \text{if X[i] } \neq \text{ Tey[i]} \end{cases}$$

where X represents the consonant in current tonality, Tey represents consonant in C major, i represents the number of notes where they located. We named this variable T as tonality changing list, and T_i as **tonality changing constant**, represent in α.

2.2 Accidental of Letters

Sometimes, some specific notes will be changed to fit the sheet. In music, an accidental is given to symbolize this alternation; accordingly, in Frasia, a variation mark will appear in the paragraphs to indicate this change. To digitalize this changing, we also follow the principle mentioned in the section 2.1, where 0 represents the changing status of a letter and 1 represents the non-changing status.

Here is an example, to spell word *Frasia* in Frasia (Tey), we have:

$$Fyv- : lor - hax - siu - min - hax \tag{5}$$

which represents as :

$$[1, 0, 1, 1, 1, 1] \tag{6}$$

in format of single changing constant.

Similarly, we can see the logic used in this transition:

$$A[i] = \begin{cases} 1, & \text{if X'[i] = X[i]} \\ 0, & \text{if X'[i] } \neq \text{ X[i]} \end{cases}$$

where X' represents the current letter's reading, X represents the basic consonant reading in current tonality, i represents the number of letters where they located. We named this variable A as the list of the single changing state, $A[i]$ as **single changing constant**, which represents in β

2.3 Sustenance in Words

Aside of the two changing types mentioned in the previous section, there is another transformation called sustenance in Frasia. Similarly, to digitalize this change, let the number 0 represent the non-sustenance status of a letter, and the number 1 represent the sustenance status. One more time, we use the word *Frasia* (in Tey) as an example, we can clearly see that there are sustenance marks in the letter r and the last a, which can be represented as:

$$[0, 1, 0, 0, 0, 1] \tag{7}$$

Again, the logic used in this transition is:

$$S[i] = \begin{cases} 1, & \text{if type}(X'[\text{i}]) = \text{type } (X'[\text{i-1}]) \\ 0, & \text{if type}(X'[\text{i}]) \neq \text{type}(X'[\text{i-1}]) \end{cases}$$

where X' represents the current letter's reading, type() refers to the type of current letter (either consonant or vowel), i represents the number of letters where they located and $i \geq 1$. We named this variable S as the list of sustenance state, $S[i]$ as **sustenance constant**, represents in ϵ. Specially, we have S(0) = 0 for each paragraph.

2.4 Vowel or Consonant?

After digitalize the 3 transition forms in the previous sections, we still have one last question to solve, that is, how do we know if the current reading letter is vowel or consonant? Basically, the type of letter is determined by 2 factors, which are the previous letter format, and the existence of sustenance mark.

Letter	i	S(i)	F(i)
F	0	0	1
r	1	1	1
a	2	0	0
s	3	0	1
i	4	0	0
a	5	1	0

Table 1: Sample of Letter Format Constant

Theoretically, a letter will be a consonant if the previous letter is a vowel and does not have a sustenance mark itself, or the previous letter is a consonant and has a sustenance mark itself. Equivalently, if a letter is a vowel, either the previous letter is a consonant and without a sustenance mark itself, or the previous letter is a vowel and with a sustenance mark itself.

This time, we assign a variable called **letter format constant** to describe the vowel in 0 and the consonant in 1, which represents in δ. From the above argument, we have the following:

$$F[i] = \begin{cases} 1, & \text{if S(i) = F(i-1)} \\ 0, & \text{if S(i) } \neq \text{ F(i-1)} \end{cases}$$

where i $\geq$ 1, which represents the number of letters location. Specially, we have F(0) = 1 for sentences of the major scale, and F(0) = 0 for sentences of the minor scale.

Take word *Frasia* in Tey again as an example, summarize the letters and their sustenance constant ϵ, and letter format constant δ in the following table 1. We can easily see that for letter r, F(1) = 1 because S(1) = F(0), and for letter a, since S(2) $\neq$ F(1), F(2) = 0, which shows it is a vowel.

3 Formatting

In this section, we will propose a specific package format specially for Frasia message, which contains the consonant discussed in section 2. Normally, when sending a packet from source A to B, we have to consider following questions: $Q1$. the total length or the size of the message, $Q2$. the construction and component of the message unit, and $Q3$. the receiver (and the sender accordingly) action of processing the message. We will follow above questions to explain our design of a basic Frasia message packet.

Before starting everything, we propose a mapping from hexadecimal numbers to each Frasia letter for easier transcoding from Frasia letters to binary numbers, as in table 2.

All the next-coming context and further analysis will follow this assignment. Note that **Consonant First** refers to the major reading format in Frasia (i.e. singing started with consonant letters), while **Vowel First** refers to the minor reading format (i.e. singing started with vowel letters).

Frasia Characters	Value
$Teyd$	0
Hax	1
Min	2
Fyv	3
$Paib$	4
Lor	5
Keg	6
Zao	7
Cei	8
Siu	9
Thh	A
Tru	B
Space / Comma	C
Enter / Period	D
Consonant First	E
Vowel First	F

Table 2: Frasia Letters in Hexa

3.1 Size

When musicians performing the music, they always know how to split the lyrics and breath naturally to avoid the disharmonious gaps which would destroy their performance. However, as scientist, we are only aware of the song as an entirety instead of the energy flow inside the music sheet. Thus, we must find a considerable way to analyze the suitable breaks between each of the Frasia sentence. In other words, our algorithm has to know how to split the whole message in a appropriate packet length to optimize the system efficiency.

Consider a sample of signal X^{L*M*N} which collecting from a song source signal, defining as a three dimensional tensor in format of:

$$X^{(r)}(l,m,n) = A(l,r)B(m,r)C(n,r) \tag{8}$$

where L is the signal channel, M is the signal spectrum, N is the time slicing number, $r \in Rank(X)$.

By applying decomposition to this tensor, we shall be able to have mode-N factorization $X_{(N)} = X^{N*LM}$, which represent the factor of the energy generated from the source in current channel per time slice. N will be the number of script pieces we target to split.

To find the optimized N, we will use PARAFAC (Parallel Factor Analysis) [3] model for calculating. According to the model, for each $x_{l,m,n} \in X$, we have:

$$x_{l,m,n} = \sum_{r=1}^{R} a_{l,r} b_{m,r} c_{n,r} + e_r \tag{9}$$

where $a \in A^{L*R}, b \in B^{M*R}, c \in C^{N*R}, R = Rank(X)$, $e \in E$ represent the residual value.

The mode-N factorization can be written as:

$$X_{(N)} = C(B \odot A)^T + E \tag{10}$$

where $\odot$ is defined as Khatri-Rao product value as:

$$A \odot B = \begin{bmatrix} BD_1(A) \\ BD_2(A) \\ ... \\ BD_j(A) \end{bmatrix} \tag{11}$$

where $D_j(A)$ is the diagonal matrix of A.

By minimizing the value of E, we can have the most preferred N, which indicate the most appropriate time interval we shall apply.

Assume the song length is L , the average sentence length (i.e. the packet size) p will be written as:

$$p = \frac{L}{N} \tag{12}$$

Noticed that p is only a supervised value for algorithm using and indicates that the time interval for each of the song's sentences is equal, which is not always true during real performances. However, musicians still can use p as a guidance to help them perform the song that they did not learn before.

The above equation 10, can also be written in:

$$X_{(N)} = TS + \epsilon \tag{13}$$

i.e.

$$X_{(N)} \approx TS \tag{14}$$

where S is the original sound source (e.g. someone's singing, artificial electrical signal, etc.), ϵ is the error value. To calculate T, we have:

$$\min_T ||X_{(N)} - TS||_2^2 \tag{15}$$

Assign $f = X_{(N)} - TS$, we have:

$$\frac{\partial f}{\partial T} = -2(X - TS)S^T \tag{16}$$

To minimize, we let 16 equals 0:

$$(X - TS)S^T = 0 \tag{17}$$

Thus:

$$T = X_{(N)} S^T (SS^T)^{-1} \tag{18}$$

T represent the potential reflection model from the original source S to our observation $X_{(N)}$, which were caused by several real factors (e.g. time, types of songs, types of musicians, etc.), named as time coefficient matrix, is what we can use to derive our own source to the target energy output. For example, we can apply the coefficient value T to the artificial source x to reproduce a known source s. T will vary based on the situation and need to be calculate accordingly.

Noted that we assume the mapping module is linear, which means it is only applicable to specific linear-liked energy releasing scenarios. The feasibility of using a linear module to analyze the time-varying low-rank system has already been approved [5]. T may not be reliable when the Frasia song energy module discontinuous or non-differentiable.

3.2 Coding

This section will discuss a possible message format structure of Frasia message packets. There are some common agreements before we started. Firstly, we recognize that the words written in **Frasia letters format** are immutable, that is, the word Frasia in Tey should be read as Frasia, but Vrashia in Ha, while the writing of "Frasia" word keeps the same as in equation 5. To be more accurate, this article will use **Frasia written format** (like equation 5) and **English written format** (like word *Frasia* and *Vrashia*) to distinguish these two different written styles.

3.2.1 Encoding and Decoding

To enhance the readability of a digital Frasia message, several functional parts should be added to the beginning of the message packet. Referring to the existing 802.1q [9] encoding scheme, we designed this Frasia packet basic factors as below:

- **Tag (8 bits):** indicates the tonality of current Frasia words. Tag should only appear at the beginning of the message, this constant is consisting of 4 bits Major/Minor indicator and 4 bits of tonality value (followed Table Frasia Letters in Hexa). For example, to write the tonality Tey, we have *E0*, where *E* represents the major and *0* (0x00) represents the letter Teyd. However, this structure should be scalable in the future, which not only has the ability to transfer Frasia letters in different tonalities, but also allows various actions happened during message transferring.

- **Length:** indicates the length of the value message, this value should be calculated at local side before sending the message. The maximum length that the receiver can proceed should depend on its own processing ability, the only limitation is that, this number should be multiples of 8 to fulfill the text format described in Algorithm 1. In this design, we propose the maximum number as 1480 (0x5C8) which constrained by current device ability. Meanwhile, this number can be derived to 2 parts as SEQ and CT for further processing, see 3.2.3.

- **Transformation Constant (4 bits):** represents the transformation status of the current Frasia letter, each transformation constant consists of the sustenance constant, the letter format constant, the tonality changing constant, and the single changing constant, and should be added in head of each digital letters.

- **Value:** the actual letters to transfer, only accept Frasia characters in the current design.

The procedure for generating the message should follow the algorithm 1 and be done locally before sending the message. One example is, to encode the word *Frasia* in Tey, we have the table 3.

Algorithm 1 Mapping from Frasia Letters to Binary

Input: s ▷ Frasia Letters with Transformation Marks
Output: χ ▷ In Hexa
  $\epsilon, \delta, \alpha, \beta \leftarrow value$ ▷ Find the value (0 or 1) according to previous rules in section 2.
  x ▷ Find current letter in Hex form based on Table Frasia Letters in Hex
  $\chi = combineString(Tonality, Length, \sum(\epsilon, \delta, \alpha, \beta, x))$
  return χ

Message (Original)	Transformation Constant	Character Value	Message (in Hex)
Tey	*NA*	0	E0
Length	*NA*	*NA*	0030
F	0111	3	73
r	1110	5	E5
a	0011	1	31
s	0111	9	79
i	0011	2	32
a	1011	1	B1

Table 3: Sample of Mapping from Letters to Binary

Thus, the message will be written as `E0003073E5317932B1` to represent word *Frasia* in Tey tonality. The input of this encoding function is the Frasia writing format, and the output is binary (hexadecimal) message format.

The logic used to decode the message packet into Frasia characters is similar to encoding. The action of mapping from binary message to Frasia letters will refer to the Table Frasia Letters in hexadecimal, which applies to the Tag and Value translation. Noticed that the decoding algorithm does not require tonality as input. The length number will act as a global calculator that allows both the local and remote side use for counting receiving words.

Mainly, for the Transformation Constant of each value, we have the algorithm 2 to check which character it should output. The Frasia consonant table mentioned in the algorithm is defined as in table 4.

Algorithm 2 Mapping from Binary to Letters

Input: $\chi \leftarrow (\epsilon, \delta, \alpha, \beta)$
Output: s ▷ The actual English written letter in string
  for $l \leq length - 1$ **do**
    if $\delta = 0$ **then**
      $s \leftarrow$ `vowel` ▷ Check local Frasia letter table
    else if $\delta = 1$ **then**
      if $\alpha = \beta$ **then**
        $s \leftarrow$ `original consonant` ▷ Check local Frasia consonant table
      else if $\alpha \neq \beta$ **then**
        $s \leftarrow$ `changed consonant` ▷ Check local Frasia consonant table
      end if
    end if
  end for
  return s

Frasia Characters	Original Consonant	Changed Consonant
Teyd	T	D
Hax	H	X
Min	M	N
Fyv	F	V
Paib	P	B
Lor	L	R
Keg	K	G
Zao	Z	Zh
Cei	C	Ch
Siu	S	Sh
Thh	Th	Dh
Tru	Tr	Dr

Table 4: Frasia Consonant

Notice that the input of this encoding function is binary message format and output is **English written format** for easy reading. The rules used to accomplish the transferring between English and Frasia written format can simply refer to the section2, by which could reduce the readability of the message at the same time.

3.2.2 Translation

To define the translation action in Frasia characters, we still follow the rules mentioned at the beginning of the section 3.2. The translation will only change the verbal reading (English written format) but not the handwriting in Frasia written format.

Recall the message format that has been discussed in section 3.2.1, there is one digit α that we are using to represent the tonality changing status. To fulfill the translation requirement, we just need to change this variable to the value they are at in the target tonality. Basically, the required input for translation is Frasia written format characters, and the output is in English written format. We should use Algorithm 1 to generate the message in binary format, then change the α value and use Algorithm 2 to translate the characters again. The algorithm 3 is the one we specially used to change the α value. Here is an example using the algorithm to translate the word Frasia from tonality Tey to Ha as in table 5.

Algorithm 3 Tonality Translation

Input: χ, C, C' ▷ where χ represents the input Hexa string, C refers to the current tonality constant, C' refers to the target tonality constant
Output: χ'
  for α `in` χ **do**
    $\alpha \leftarrow \alpha'$ ▷ α' from C'
  end for
  $\chi' = combineString((\epsilon, \delta, \alpha', \beta), x)$
  return χ'

Message Info	Transformation Constant in Tey	Transformation Constant in Ha	Results
F	0111	0101	V
r	1110	1110	r
a	0011	0011	a
s	0111	0101	sh
i	0011	0011	i
a	1011	1011	a

Table 5: Sample of Translation from Tey to Ha

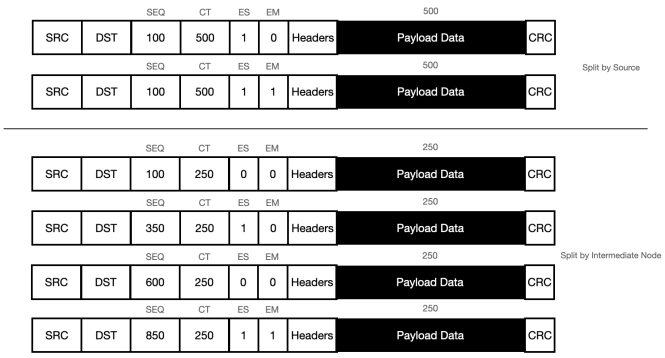

Figure 2: Message splitting and packet splitting

3.2.3 Segment

As the message length is controlled by the processing ability of all the devices on the line, we expect it will become varied and need adjustment among times. Thus, segment and reassembly techniques should be applied where applicable. For example, The sequencing and reassembly mechanism proposed by TCP protocol [4] can be applied to the algorithm 2 for further optimization, which requires an extra functional head-end for the message packet.

In TCP protocol, we have 1 bit ES (End of Segment) and 1 bit EM (End of Message) flag to represent the message status and SEQ (Sequence Number Field, indicate the relative location of the first byte of the message) and CT (Byte Count Filed, count for the length of text in current message but exclude checksum) to represent the message starting index and end position (equals to $SEQ+CT$) for re-allocation. $ES = 1$ means this packet is the end of the segment, and $EM = 1$ means this segment is the end of the message as shown in figure 2 [4].

Similarly, in Frasia decoding, we can also introduce EM, ES, SEQ and CT flags to our packet to expand our message length. All the flags will be inserted at the sender side, and ES value will be adjusted during the transmission of intermediate node, while EM value should keep constantly. Checksum will be calculated at the sender side and be verified at receiver side per each segment.

3.3 Security Addressing

Aside from the transformation constant, the initiator and receiver are both important in the Frasia message. For security and management concerns, this section will propose

Figure 3: Addressing Design

a globally permanent and unique address code design for the objects and individuals involved in the Frasia message in the figure 3. Basically, the address code is a 48-bits-long (i.e. 12 digits) hexadecimal message which consists of (list in sequence):

- **Country Code** (8 bits/ 2 digits) to represent an independent country in the world. Note that the 00 is used for non-governmental organization and FF for broadcast. In total, 254 country codes are supported in this design.
- **Province or Institution Code** (8 bits/ 2 digits) to represent the province or institution in a certain country region. Note that 00 is for non-specified area and FF is for broadcast. In total, 254 province or institution codes are supported per country in this design.
- **Registration Code** (8 bits/ 2 digits) to represent a certain functional song (lyrics) which has already been officially registered. These codes are designed to fasten the processing speed. Note that 00 is for non-registered song and FF is for broadcast. In total, 254 unique song codes are supported per individual or objects in this design.
- **Personalized Identity Code** (24 bits/ 6 digits) to represent a unique identity code of an individual or an object. Note that 0x00 is for non-registered object and FxFF is for broadcast. In total, 16777216 identity codes are supported per province or institution.

The limitation of these designs is quite visible, we should consider the possibilities of extension of personalized identity code, as the time is coming and the population of musicians will significantly increase for sure.

3.4 Mechanism

In order to transfer the Frasia message, we designed the mechanism of each sector of the packet to let both side of the communication knows how to identify and respond the packets in this section. Notice that our mechanism only applies to the P2P scenario, the P2MP mechanism and its corresponding packet format will not be considered.

The workflow at sender side is proposed as:

1. Sender transforms all the Frasia letters to binary payload by using algorithm 2 and slice the payload to segments if needed. We shall expect a specifically designed inter-node between the sender and the receiver to take response of transferring and processing the slicing and segment in the future [4].
2. Sender calculates the CRC code according to the payload results in step 1 and add to the end of the payload (per segment).
3. Sender adds Tag (Tonality) and length information at the beginning of the each packets.
4. Sender calculates the necessary segment header and add to the beginning of the packets.
5. Sender adds the address header to the beginning of the packets.

The total workflow at receiver side is proposed as below:

1. Receiver compared the address in the message packet and its own table, and drop the message if mismatching.
2. Receiver check the Tag value in the header and choose the corresponding tonality.
3. Receiver use CRC to check the completeness of the segment. If check passed, use algorithm 4 to recombine the message.
4. Receiver responds one special stopping message to the sender to end the conversation.

Algorithm 4 Recombination

Input: $F : a, s, l, t, p, CRC$ ▷ Message packet: address header, segment header, length, tag, payload, CRC
Output: F' ▷ Frasia message in English written format

```
pointer = SEQ
message = [ ]
while EM ≠ True do
    for pointer ≤ l − 1 do
        sentence = [ ]
        while ES ≠ True do
            p' ← s ▷ Return of Algorithm 2 as input = p
            sentence[pointer].append(p')
            pointer += CT
        end while
        message.append(sentence)
    end for
end while
return message
```

Basically, a standard Frasia message will have a format as below:

- **Address Header** Contains the address of receiver and the initiator, which can be either an object or an individual, the addressing format should refer to the section 3.3. Security check will be performed once packet received, the receiver should drop message if the value does not match locally.
- **Segment Header** Contains SEQ, CT, ES, EM value, after the node receive the message, it will act accordingly, see section 3.2.3.

Figure 4: Standard Frasia Message Packet Sample

- **Tag** refers to the tonality of the carried message, see section 3.2.
- **Payload Data** refers to the actual message carried by the data packets, for example, the lyrics. The sender will use algorithm 1 to encode data, and the receiver will use algorithm 2 to decode data.
- **CRC** (Cyclic Redundancy Check) refers to a commonly used algorithm in existing networks to check for bit error[15].

A standalone Frasia message packet should appear as in figure 4.

4 Frasia System

In this section, we will analysis the 1544 case [2], which is one of the largest M-energy explosion we can observe at current time by using the equations and design principles discussed in section 3, through which to propose a general structure of a complex Frasia system and its chiral version, and by which to can have an idea of how to achieve the completion for large-scaled songs.

Before starting over, we have to clarify some definition as below. **Complex Frasia Network**: meaning the spell-casting system involved more than one person, or the spell-casting system can be separated independently to different children systems. This structure is capable to carry multiple tonalities and allows different songs running simultaneously **Chiral Version**: meaning the energy produced via this chiral system can be completely offset by its original system and vise versa.

4.1 Background Analysis

Trace back the case that occurred in December 3Epi.1544 [2], where a single summon song ruined St.Godhrkar Theological Seminary. From the recording, we can easily see that the accident was composed by following 3 parts:

- **Shervlett** was the person who sang the summon song of the Ychis. Obviously, she was responsible for the main function (i.e. summon) in this "concert", and passed her rhythm to the Chorus.
- **St.Godhrkar Chorus** was affected at the same time, they started the vocal accompaniment for Shervlett, which provided a steady energy field for Shervlett to sing.
- **Symphony Orchestra** responded to Shervlett's rhythm and started playing with their special instruments, which expanded the scope of influence due to the material used. The orchestra enhanced the rhythm from Shervlett and "helped" her to break through the wall of time and space.

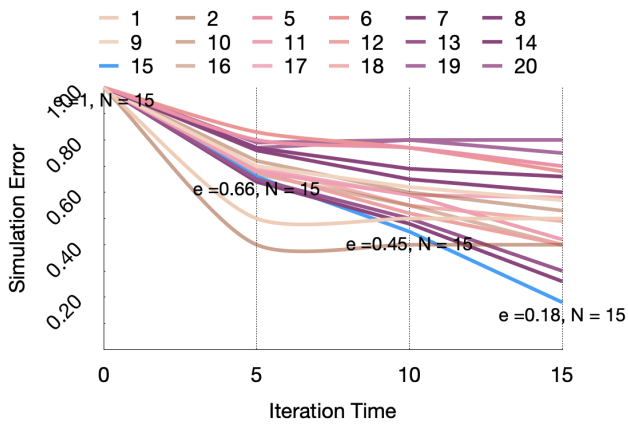

Figure 5: Simulation Results of 1544 Case

In this case, by applying the equation 8, we have:

- $L = 6$, as the data provided by Ychis is in 6-dimensioned.
- $M = M_{(t)}$, which is a time varied matrix that describes the current energy reading among dimensions.
- $Rank = 3$, as we have 3 voice parts.

where the PARAFAC model [3] can be used to decompose the tensor collected from the Y.Godhrkar's record to find the best N. The results in the figure 5 shows that when $N = 15$, the decomposing is the best simulation. The full script is in the Appendix 7. Obviously, this case is a Frasia complex system.

Evaluation The script has been tested and verified by the musicians from *Recording Convergence Group* on 1293572th standard day, Co-Lab members may also access the full script through *The Ring Library*.

4.2 Experiment

In this section, we are going to generate the opposite song of the 1544 song we discussed in section 4.1 through our self-designed system. We expect this opposite song is the chiral system that generates the opposite energy which is able to counterweigh the effect of the 1544 song, while has the similar structure of the complex system it refers to.

Theoretically, to find the opposite song (i.e. reverse wave), we need to have:

$$X^{inv} = -X \tag{19}$$

which can also be written as:

$$x^{inv}_{l,m,n} \approx \sum_{r=1}^{R} a_{l,r} b_{m,r} (-c_{n,r}) \tag{20}$$

As we already known, mode-c decomposition as equation 14, we need to know T^{inv} where:

$$X^{inv}_{N} \approx T^{inv} S \tag{21}$$

By minimizing:

$$\min ||X + T^{inv}S|| \tag{22}$$

we have:

$$T^{inv} = -X_{(N)}S^T(SS^T)^{-1} \tag{23}$$

Recall the equation 18, we have:

$$T^{inv} = -T \tag{24}$$

Thus, once we are aware of $X_{(N)}$ and T, we shall able to find the reverse wave X^{inv}, named as η wave.

However, as limited by the resource of *The Ring*, we shall find a way to **1.** separate the whole performance in small sections as our M-energy generator is restricted, and **2.** transfer the Frasia message efficiently as the time window for *The Ring's* channel to Co.Lab is tight. The following sections will discuss these 2 problem.

4.2.1 Structure

To improve the efficiency of energy transmitting and save the calculation resources, we re-designed the whole structure of the song which allows several musicians perform synchronically. Motivated by the Frasia system we discussed in 4.1 and the design of satellite network system [6], we explained our experimental structure as following 3 parts, as figure 6:

Definition (Shervlett): This is the part that defines the core functions of the whole system, like the motif we mentioned in the previous paragraph. The tonality used in this component depends on the function that the system is going to achieve, which can be **Teyd, Paib, Zaozh**, etc. This part is only responsible for generating and proposing the core function of the system and will not be responsible to respond to the users from the outside environment. However, it still needs to retrieve data and write results in the Auxiliary part.

Auxiliary (Symphony Orchestra): Like the music sheet in a concert, this is the part where all the basic music tone and Frasia transformation tables stored, and related temporary calculation results from message generation are located, which acts as a storage pool. The system itself can read and write the information to this component, while the outside users do not interact with this part of the system at all. The process of data transferring between each layer can be viewed as cognitive interference with the individual or object's mental field. Thus, this layer is where most of the **Fyv** tonality message work. Automatic machines may also be involved in this layer to create precise and steady music tones.

Multiplex (St.Godhrkar Chorus): Multiplex layer will retrieve the results generated by the Definition layer from the Auxiliary layer records, and amplify to fulfill the requirements either from outside or inside. This is also where the interaction between inside and out happens, it will process the outside users' command (if any) and pass the desired results to the Application layer. Consider that the procedure of adjusting and alignment of the original data is making an influence on the rhythm, **Hax** tonality should be quite suitable for this layer's.

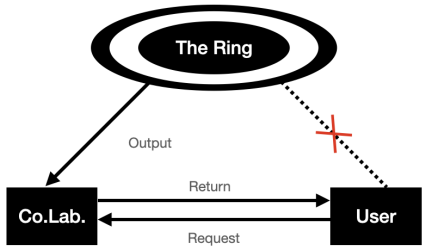

Figure 6: The System Structure

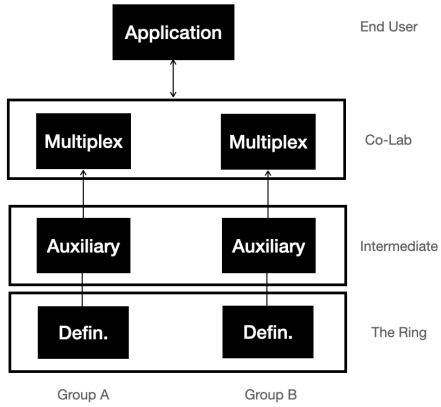

Figure 7: Basic Frasia Network Structure (size:2)

To fulfill the requirements of easy expansion and reduce the link between *The Ring* and end users, we refer to the Fat-Tree [7] [12] network to build our structure. In summary, A Frasia network will have k pods (i.e. the song was literally divided into k parts, $N = k$), each k pod has 2 layers of performing component, and 1 layer of output to further interact with the outside environment. Each component will be connected to $\frac{2}{k}$ components from other layers, and the total number of the connection of one component is k.

Figure 7 is an example of the basic Frasia network in size 2. The *Application* layer represents the interaction between external users and this Frasia network system, the results from output, and the commands from users (if any) will be processed. The detailed working mechanism of this part will not be discussed here.

Notice that this structure assume that all the individuals or objects involved are considered as fully capable to complete the task assigned, and there is no energy nor signal interference or loss happened during the transportation inside the system, or the system is strong enough to fix the loss itself. As for now, the operators from the *Recording Convergence Group* are able to fulfill these requirements. However, we should expect the redundancy part specifically designed for the whole structure in the future.

4.2.2 Isolation

To allow multiple tonality simultaneously exist inside the same message as the messages transferred between the def-

Figure 8: Two-Layered Frasia Message

inition and auxiliary layer will contain at least 2 different tonality as designed in section 4.2.1, we have to design a protocol that can show the lyrics in the sequence, as well as indicates the target tonality of each part. By adjusting the message format mentioned in the section 3.4, we have what in the figure 8

The definition of the new adding part amended from 802.1ad packets [8] is as follows :

- **Flag** (16 bits each) to indicate this message is specially used for multilayer Frasia message. If the receiver does not support this kind of message, it should drop the receiving packets. The default value is set to 0x8100.
- **Tag(n)** (8 bits each) to show the tonality of each part of the data. Note that if the expected results are in English written format only, the decode of the Tag is not necessary, since the transformation constant is able to describe the letters already. These Tags are used for further mapping like music sheet or text recording.
- **NID** (Network Identity, 4 bits each) to show the length of each specific payload. The sum of the NID value should be equal to the length of the full payload data.

This design allows system to transmit multiple tonalities more efficiently, as we lack the link resource between *The Ring* and Co.Lab. We can save more space for messages when their source and destination are the same. The design of Flag-Tag-NID could also be expanded to more than 2 layers, which increase the complexity and flexibility of the message, but will reduce the efficiency to transfer the actual data at the same time. More studies may be required on this problem.

4.2.3 Results

η wave has been successfully generated at around 5:am on the 44165th standard day. Records can be retrieved from *The Ring Library* [10]. The residual value after mix the 1544 case song and the η wave is at 0.000031, which is less than the background radiation of any Epi., meaning the effect of the 1544 song can be absorbed naturally in 7 days after η wave applied.

4.2.4 Related Work

Some work has been done[17] to prove that the Bayes error can improve the certified accuracy to 67.49% of the system output when adversarial samples has been introduced to the complex system to enhance the system robustness.

5 Conclusion

This paper presents a binary simulation model of Frasia words and sentences with their transformation status. Based on this model, we design an formatted protocol with several special components to transfer Frasia words through electric signals. We found that this design can be expanded to establish as a whole system which allows the co-working and distinguishing of different tonalities' song without having obstacles. Furthermore, we found that this system structure can transcribe the 3Epi.1544 case song energy and its opposite one, which shows its efficiency in analyzing and processing M-energy.

6 Future Study

In section 4.1, we discussed the best slicing number N of the song and we have noticed that when $N = 1$ and $N = 2$, the efficiency of performing is better than any other assumption N when iteration time less than 10, and better than 60% of other N assumption at iteration time reaches 15. Thus, we shall consider $N = 1$ or $N = 2$ still as a considerable optimization of the song slicing. In other words, further study needs to be done at the fields which help to compose or reduce the song script in order to be performed as an entity. Related work has been done in reference [13].

7 History Report

We have observed η wave on 5:am 1293578th standard day, i.e. 12th December 3Epi. 1544, after section 4.2 finished.

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Appendix

Device List

All the devices involved in section 4.2 are listed in table 6.

Song Script

Fua nie zhogta ionetine stolie
  Gunfeg ozhia(x)n pan syala si niecie
  Fsaieni fa hsi iun f(x)lasialie
  Ileliteg pallayan iun hiesitie
  Puo li, puo li, puo li — fagyulanute
  Lela sunan sunan pin gupang
  Flasia, flasia, flasia — fua ni fahsi
  Ju nelustan ailisaltinanie
  Foila, foila, foila — ein lapasang

Name	Location	Function
RR1-RR4/ Colo-A/ Sec-A/CORE	The Ring	Core function calculation (Definition)
AUX1-AUX3/ Colo-A /Sec-B/COMM	The Ring	Signal amplification, responding to the ground (Multiplex)
RP1/Lab /Sec-31/COMM	Co.Lab	Signal amplification, respond to The Ring and users (Multiplex)
RP2-RP3/Lab /Sec-31/COMM	Co.Lab	Perform end-users' request

Table 6: Test Device Involved

Hiestin, g(x)ailise pan patesa nyala

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